

Matius Robert

matrobert@outlook.com | mtrob.io

 [mtrob](#) |  [matrobert](#) |  [matrobert_](#)

Singapore

EXPERIENCE

- **Doctoral Research (PhD)** Aug 2025 - Current
Cancer and Stem Cell Biology Signature Research Programme, Duke-NUS Medical School Singapore
 - Drive multi-year pre-clinical research projects to uncover novel targets that underlies the mechanisms of aging, metabolic dysfunction, and carcinogenesis.
- **Research Assistant** Sept 2021 - May 2025
NUHS Centre for Healthy Longevity Singapore
 - Spearheaded research projects and collaborations investigating the senescence secretome and extrachromosomal DNA (ecDNA) in cancer progression.
 - Secured research funding via the HLTRP Young Investigator Grant to launch a new pilot study on extrachromosomal DNA.
 - Designed and executed differential gene expression analysis and clinical data mining on large-scale public datasets (TCGA, TCIA, and single-cell atlases) using R.
 - Techniques implemented include time-lapse confocal microscopy, karyotyping, fluorescence in-situ hybridization (FISH), and DNA content profiling via flow cytometry.
- **Undergraduate Researcher** Aug 2020 - Jun 2021
Department of Physiology, National University of Singapore Singapore
 - Designed and executed a loss-of-function study to investigate the role of a histone deacetylase for specific targeting of glioblastoma cancer stem cells.
 - Techniques implemented include site-directed mutagenesis, limiting dilution assay, and in-vitro cancer phenotypic assays.
- **Undergraduate Researcher** June 2019 - April 2020
Department of Biological Sciences, National University of Singapore Singapore
 - Carried out phenotypic & molecular characterization of a plant disease resistance protein in the context of stomatal development in Arabidopsis.
 - Implemented protein-protein interaction assays such as yeast two hybrid and bimolecular fluorescence complementation for validation of high-throughput screening hits.

EDUCATION

- **Doctor of Philosophy** 2025 - 2029
Duke-NUS Medical School Singapore
 - Recipient of Duke-NUS PhD Fellowship
- **Bachelor of Science (Hons.)** 2017 - 2021
National University of Singapore Singapore
 - Recipient of the Student Life Award (Community Engagement - Merit)
 - Societies: NUSSU Committee for Information Technology; NUS Angklung Ensemble; NUS Cultural Activities Club

PUBLICATIONS

J=Journal, P=Pre-print

- [J.1] **M. Robert**, B.K. Kennedy, K.C. Crasta. (2024). **Therapy-induced senescence through the redox lens**. *Redox Biology*, Vol. 74, Article 103228. DOI: 10.1016/j.redox.2024.103228
- [J.2] Y.H Chuah, E.X.Y Tay, O.V. Grinchuk, J. Yoon, J. Feng, S. Kannan, **M. Robert**, R. Jakhar, Y. Liang, B.W.L Lee, L.C. Wang, Y.T. Lim, T. Zhao, R.M. Sobota, G. Lu, B.C. Low, K.C. Crasta, C.S. Verma, Z. Lin, D.S.T Ong. (2023). **CAMK2D serves as a molecular scaffold for RNF8-MAD2 complex to induce mitotic checkpoint in glioma**. *Cell Death & Differentiation*, Vol. 30, pp. 1973-1987. DOI: 10.1038/s41418-023-01192-3
- [J.3] **M. Robert**, K.C. Crasta. (2022). **Breaking the vicious circle: extrachromosomal circular DNA as an emerging player in tumour evolution**. *Seminars in Cell & Developmental Biology*, Vol. 123, pp. 140-150. DOI: 10.1016/j.semcdb.2021.11.015
- [P.1] **M. Robert***, R. Jakhar*, B.V. Au, G.W.L. Eng, M. Wei, A.K. Gao, Y.H. Chuah, K. Sachaphibulkij, I. Bonne, K.J. Lim, I. Sinha, D.B.L Teh, L.H.K. Lim, P. Kumar, N.K. Verma, N.S.K. Sze, E.H. Lim, D.S.T. Ong, J.K. Cheong, K. Itahana, J.E. Connolly, K.C. Crasta. (2024). **Tumour-derived extracellular vesicles within the therapy-induced senescent secretome distinctly suppress breast cancer via DKK1-mediated inflammatory response**. *BioRxiv*. DOI: 10.1101/2024.03.29.586905

CONFERENCES & TALKS

O=Oral, PO=Poster

- [O.1] **Matius Robert***, Rekha Jakhar*, Bijin Veonice Au, Gracie Wee Ling Eng, Alvin Kunyao Gao, You Heng Chuah, Karishma Sachaphibulkij, Lina Hsiu Kim Lim, Derrick Sek Tong Ong, Elaine Hsuen Lim, Yoko Itahana, Koji Itahana, Jit Kong Cheong, John Edward Connolly, Karen Carmelina Crasta. (2024). **Extracellular vesicles within a tumourigenic senescent tumour secretome are able to confer antitumour immunity**. In *N2CR Early Cancer Intervention Symposium*, 13-14 May 2024, MD6 Building, National University of Singapore, Singapore.
- [PO.1] **Matius Robert***, Rekha Jakhar*, Bijin Veonice Au, Gracie Wee Ling Eng, Alvin Kunyao Gao, You Heng Chuah, Karishma Sachaphibulkij, Lina Hsiu Kim Lim, Derrick Sek Tong Ong, Elaine Hsuen Lim, Yoko Itahana, Koji Itahana, Jit Kong Cheong, John Edward Connolly, Karen Carmelina Crasta. (2023). **Small Extracellular Vesicles from Therapy-induced Senescent TNBC Elicit Paracrine Anti-tumour Effects**. In *Frontiers in Cancer Science*, 6-8 November 2023, University Cultural Centre, National University of Singapore, Singapore.
- [PO.2] **Matius Robert***, Rekha Jakhar*, Karen Carmelina Crasta. (2022). **Role of small extracellular vesicles in mediating paracrine activities of therapy-induced senescent breast cancer**. In *Longevity Science Singapore Conference*, 7-8 September 2022, NUHS Tower Block, National University of Singapore, Singapore.

SKILLS

- **Programming Languages:** R, Python, Bash
- **Version Control & Deployment:** Git, Docker
- **Specialized Area & Interest:** Senescence, aging, metabolic diseases, genomic instability, multi-omics
- **Languages:** English (Bilingual), Bahasa Indonesia (Native)

HONORS AND AWARDS

- **Duke-NUS PhD Fellowship** Aug 2025
Duke-NUS Medical School 
 - Duke-NUS PhD Fellowship is a merit scholarship awarded to graduate students under the Duke-NUS PhD programme. The scholarship provides monthly stipend, travel grant, tuition fee waiver, and other miscellaneous student fee subsidy.
- **HLTRP Young Investigator Grant** Nov 2024
Healthy Longevity Translational Research Programme 
 - HLTRP Young Investigator Grant aims to support innovation amongst junior scientists. The grant awards selected research proposals with emphasis on blue-sky research and novelty in the field of healthy longevity.
- **The Student Life Awards (Community Engagement - Merit)** Oct 2019
NUS Office of Student Affairs 
 - The Student Life Awards is an annual event organized by the Office of Student Affairs (OSA) that aims to recognize exceptional students for their noteworthy contributions to the University outside of academia.

VOLUNTEER EXPERIENCE

- **Coordinator & Volunteer**

May 2022 - Jan 2025

Healthy Longevity Webinar Series



- Coordinated a webinar series aimed at raising awareness of healthy aging, featuring leading scientists and policymakers.
- Engaged an international viewership with an average of 350 attendees per episode.
- Delivered concise presentations to translate recent geroscience findings and ongoing research for a lay audience.
- Oversaw the recruitment, scheduling, and coordination of internal presenters and moderators to ensure high-quality production workflows.

- **Volunteer**

Sept 2022

Longevity Science Singapore Conference



- Facilitated the Longevity Science Singapore Conference by providing hospitality support for international guest speakers and VIPs.
- Executed ad-hoc logistical duties to ensure the smooth operation of the full-day scientific event.

- **Logistics Director**

Feb 2018 - Oct 2018

NUANSA Cultural Production



- Directed logistics for a large-scale cultural production involving 105 students and an audience of 700, with proceeds donated to earthquake relief efforts in Palu, Indonesia.
- Led and managed the logistics team to ensure timely inventory management and operational support.
- Oversaw critical requirements, which included venue arrangements, transportation, procurement, and manpower allocation.

REFERENCES

- References available upon request